

Asymptotic Analysis

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Using Basic Methods

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O-Notation

If $f(n)$ is a growth function for an algorithm, and $g(n)$ is a function such that for **some positive real constant c** and for **all $n \geq n_0$** ,

$$0 < f(n) \leq c \cdot g(n)$$

then we say $f(n) = O(g(n))$ (Read $f(n)$ is **Big-Oh** of $g(n)$)

- The behavior of $f(n)$ and $g(n)$ is portrayed in the graph. It follows, that for $n < n_0$, $f(n)$ may be **above** or **below** $g(n)$, but **for all $n \geq n_0$, $f(n)$ falls consistently below $g(n)$** . The function $g(n)$ is said to be **asymptotic upper bound** for $f(n)$



- There can be a **several functions** for which the **$g(n)$ is the upper bound**. All such functions are said to belong to the group identified by $g(n)$. Symbolically, we denote the relationship as $f(n) \in O(g(n))$

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O-Notation

Example(1): $3n^2 + 10n \in O(n^2)$

$$10 \leq n \quad \text{for } n \geq 10 \quad (\text{obvious!})$$

$$10n \leq n^2 \quad \text{for } n \geq 10 \quad (\text{Multiplying both sides with } n)$$

$$3n^2 + 10n \leq 4n^2 \quad \text{for } n \geq 10 \quad (\text{Adding } 3n^2 \text{ to both sides})$$

$$3n^2 + 10n \leq c \cdot n^2 \quad \text{for } n \geq n_0 \quad (c=4 \text{ and } n_0=10)$$

Therefore, $3n^2 + 10n \in O(n^2)$

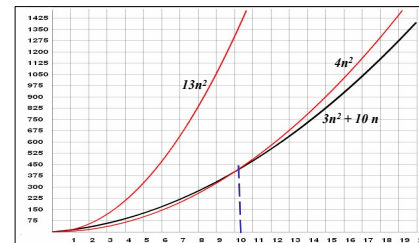
- The **choice of c is not unique**. We can select **some other c** and **corresponding n_0** so that the relation holds true. For,

$$3n^2 + 10n \leq 3n^2 + 10n^2 \quad \text{for } n \geq 1 \quad (\text{replacing } n \text{ with } n^2 \text{ on the right side})$$

$$= 13n^2 \quad \text{for } n \geq 1$$

$$3n^2 + 10n \leq c \cdot n^2 \quad \text{for } n \geq n_0 \quad (c=13 \text{ and } n_0=1)$$

- The graph depicts the two solutions. Observe that both $13n^2$ and $4n^2$ eventually grow faster than $3n^2 + 10$



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O- Notation

Example(2): The preceding technique can be applied to prove the Big-oh behavior of of polynomials. For instance, $7n^3 - 5n^2 - 20n - 16 \in O(n^3)$. Because,

$$7n^3 - 5n^2 - 20n - 16 \leq 7n^3 + 5n^2 + 20n + 16 \quad \text{for } n \geq 1 \quad (\text{changing negative terms to positive})$$

$$\leq 7n^3 + 5n^3 + 20n^3 + 16n^3 \quad \text{for } n \geq 1 \quad (\text{replacing } n^2, n, \text{ and } 1 \text{ with } n^3)$$

$$= (7+5+20+16) n^3$$

$$= 48n^3 \quad \text{for } n \geq 1$$

$$7n^3 - 5n^2 - 20n - 16 \leq c \cdot n^3 \quad \text{for } n \geq n_0 \quad (c=48, n_0=1)$$

Therefore, $7n^3 - 5n^2 - 20n - 16 \in O(n^3)$

Example(3): In general, $a_0 + a_1n + a_2n^2 + \dots + a_kn^k \in O(n^k)$, $a_k > 0$

$$\text{For, } a_0 + a_1n + a_2n^2 + \dots + a_kn^k \leq |a_0| + |a_1|n + |a_2|n^2 + \dots + |a_k|n^k \quad \text{for } n \geq 1$$

$$\leq |a_0|n^k + |a_1|n^k + |a_2|n^k + \dots + |a_k|n^k \quad \text{for } n \geq 1$$

$$= (|a_0| + |a_1| + |a_2| + \dots + |a_k|) n^k$$

$$= c \cdot n^k, \quad n \geq n_0$$

$$\text{where } c = |a_0| + |a_1| + |a_2| + \dots + |a_k| \text{ and } n_0 = 1$$

Therefore, $a_0 + a_1n + a_2n^2 + \dots + a_kn^k \in O(n^k)$

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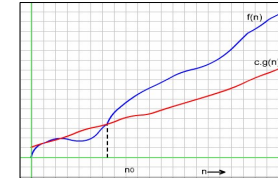
Ω -Notation

If $f(n)$ is a growth function for an algorithm, and $g(n)$ is a function such that for **some positive real constant c** and for **all $n \geq n_0$** ,

$$0 < c \cdot g(n) \leq f(n)$$

then we say $f(n) = \Omega(g(n))$ (Read $f(n)$ is **Big-Omega of $g(n)$**)

▪ The behavior of $f(n)$ and $g(n)$ is portrayed in the graph. It follows, that for $n < n_0$, $f(n)$ may be **above** or **below** $g(n)$, but **for all $n \geq n_0$, $f(n)$ falls consistently above $g(n)$** . The function $g(n)$ is said to be **asymptotic lower bound** for $f(n)$



▪ There can be a **several functions** for which the $g(n)$ is the lower bound. All such functions are said to belong to the group identified by $g(n)$. Symbolically, we denote the relationship as $f(n) \in \Omega(g(n))$

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Ω -Notation

Example(1): $n^2 - 10n \in \Omega(n^2)$.

$$n \geq n/2 \quad \text{for } n \geq 1 \quad (\text{Obvious !})$$

$$n - 10 \geq n / (2 \times 10) \quad \text{for } n \geq 10 \quad (\text{Divide right side by 10})$$

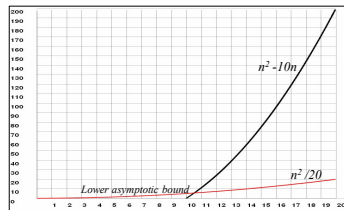
$$= n/20$$

$$n^2 - 10n \geq n^2/20 \quad \text{for } n \geq 10 \quad (\text{Multiply both sides with } n \text{ to maintain inequality})$$

$$n^2 - 10n \geq c \cdot n^2 \quad \text{for } n \geq n_0, \text{ where } c = 1/20 \text{ and } n_0 = 10$$

Therefore, $n^2 - 10n \in \Omega(n^2)$.

The behavior of functions $n^2 - 10n$ and $n^2/20$ is shown in the graph. Observe that for $n \geq 10$, the function $n^2/20$ falls below the function $n^2 - 10n$



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Ω -Notation

Example(2): $3n^2 - 25n \in \Omega(n^2)$.

$$n \geq n/2 \quad \text{for } n \geq 1 \quad (\text{Obvious !})$$

$$n - 25/3 \geq 3n / (2 \times 25) \quad \text{for } n \geq 9 \quad (\text{Divide right side by } 25/3 \approx 8.3)$$

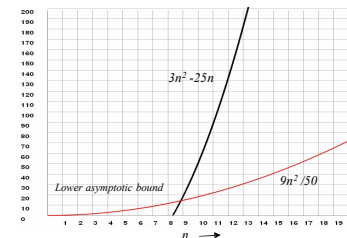
$$= 3n/50 \quad \text{for } n \geq 9$$

$$3n^2 - 25n \geq 9n^2/50 \quad \text{for } n \geq 9 \quad (\text{Multiply both sides with } 3n \text{ to maintain inequality})$$

$$3n^2 - 25n \geq c \cdot n^2 \quad \text{for } n \geq n_0, \text{ where } c = 9/50 \text{ and } n_0 = 9$$

Therefore, $3n^2 - 25n \in \Omega(n^2)$.

The behavior of functions $3n^2 - 25n$ and $9n^2/50$ is shown in the graph. Observe that for $n \geq 9$, the function $9n^2/50$ falls below the function $3n^2 - 25n$



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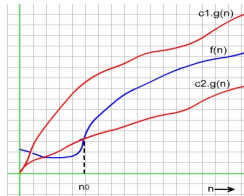
Θ -Notation

If $f(n)$ is a growth function for an algorithm, and $g(n)$ is a function such that for **some positive real constant** c_1, c_2 and for **all** $n \geq n_0$,

$$0 < c_2 g(n) \leq f(n) \leq c_1 g(n)$$

then we say $f(n) = \theta(g(n))$ (Read $f(n)$ is *theta* of $g(n)$)

- The behavior of $f(n)$ and $g(n)$ is portrayed in the graph. It follows, that for $n < n_0$, $f(n)$ may be *above* or *below* $g(n)$, but for **all** $n \geq n_0$, $f(n)$ falls consistently between $c_1 g(n)$ and $c_2 g(n)$. Function $g(n)$ is said to be the **asymptotic tight bound** for $f(n)$



- There can be a **several functions** for which the $g(n)$ is **asymptotic tight bound**. All such functions are said to belong to the group identified by $g(n)$. Symbolically, we denote the relationship as $f(n) \in \theta(g(n))$

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Θ -Notation

Example: $5n^2 - 19n \in \theta(n^2)$.

Considering the **upper bound**,

$$5n^2 - 19n \leq 5n^2 \text{ for } n \geq 0$$

$$5n^2 - 19n \leq c_1 n^2 \text{ for } n \geq n_1, \text{ where } c_1 = 5 \text{ and } n_1 = 0$$

Next, considering **lower bound**,

$$n \geq n/2 \text{ for } n \geq 1 \quad (\text{Obvious!})$$

$$n - 19/5 \geq 5n / (2 \times 19) \text{ for } n \geq 4 \quad (\text{Divide right side by } 19/5 = 3.8)$$

$$= 5n / 38 \text{ for } n \geq 4$$

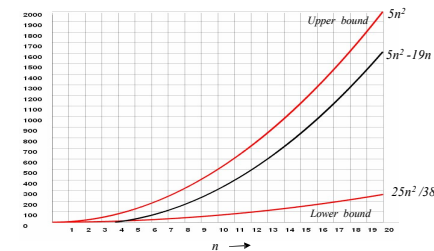
$$5n^2 - 19n \geq 25n^2 / 38 \text{ for } n \geq 4 \quad (\text{Multiply both sides with } 5n)$$

$$5n^2 - 19n \geq c_2 n^2 \text{ for } n \geq n_2, \text{ where } c_2 = 25/38 \text{ and } n_2 = 4$$

It follows, $0 < c_2 n^2 \leq 5n^2 - 19n \leq c_1 n^2$ for $n \geq n_0$, where $n_0 = 4$, $c_1 = 5$ and $c_2 = 25/38$.

Therefore, $5n^2 - 19n \in \theta(n^2)$.

- The upper and lower bounds of $5n^2 - 19n$ are shown in the graph



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o -Notation

Small oh

If $f(n)$ is a growth function for an algorithm and $g(n)$ is some function such that

$$0 \leq f(n) \leq c g(n) \text{ for all } c > 0 \text{ and all } n \geq n_0$$

then we say that

$$f(n) = o(g(n)) \quad (\text{Read } f(n) \text{ is small-oh of } g(n))$$

Symbolically, $f(n) \in o(g(n))$

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o -Notation

Example(1): $n \in o(n^2)$

Consider, $n < c n^2$ where c is **any** real constant

$$1 < c n \quad (\text{Dividing both side with } n)$$

Or,

$$n > 1/c$$

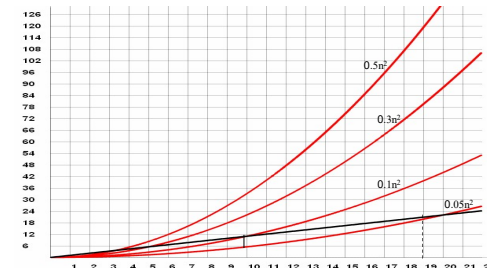
It follows that for all $c > 0$, we can find n_0 such that for all $n \geq n_0$, the above inequality holds true. For example, the **some selections of c and corresponding n_0** are

$c = 0.5$ then $n_0 = 3$; $c = 0.3$, then $n_0 = 5$; $c = 0.1$ then $n_0 = 10$; $c = 0.05$, then $n_0 = 20$

These possibilities are depicted in the graph.

It follows that for all c we can find $n \geq n_0$ such that $n \leq c n^2$

Therefore, $n \in o(n^2)$



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ω -Notation

Small omega

If $f(n)$ is a growth function for an algorithm and $g(n)$ is some function such that

$$f(n) > c \cdot g(n) \text{ for all } c > 0 \text{ and } n \geq n_0$$

then we say that

$$f(n) = \omega(g(n)) \text{ (Read } f(n) \text{ is small omega of } g(n) \text{)}$$

$$\text{Symbolically, } f(n) \in \omega(g(n))$$

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Using Limit

O -Notation

If $f(n)$ and $g(n)$ are growth functions such that

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = c, \text{ where } 0 \leq c < \infty$$

then $f(n) \in O(g(n))$.

Example(1): $3n^2 + 5n + 20 \in O(n^2)$

$$\lim_{n \rightarrow \infty} \frac{3n^2 + 5n + 20}{n^2} = 3 + 5/n + 20/n^2 = 3$$

Therefore, $3n^2 + 5n + 20 \in O(n^2)$

Example(2): $10n^2 + 25n + 7 \in O(n^3)$

$$\lim_{n \rightarrow \infty} \frac{10n^2 + 25n + 7}{n^3} = 10/n + 25/n^2 + 7/n^3 = 0$$

Therefore, $10n^2 + 25n + 7 \in O(n^3)$

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L' Hopital Rule

If $f(n)$ and $g(n)$ are growth functions such that

$$\lim_{n \rightarrow \infty} f(n)/g(n) = \infty/\infty$$

then by **L'Hopital's Rule**

$$\lim_{n \rightarrow \infty} f(n)/g(n) = \lim_{n \rightarrow \infty} f'(n)/g'(n)$$

Example: Let $f(n) = n^3$ and $g(n) = 2^n$

$$\lim_{n \rightarrow \infty} \frac{n^3}{2^n} = \infty / \infty$$

Differentiating n^3 and 2^n :

$$\lim_{n \rightarrow \infty} \frac{3n^2}{2^n \ln(2)} = \infty / \infty$$

Again differentiating $3n^2$ and $2^n \ln(2)$

$$\lim_{n \rightarrow \infty} \frac{6n}{2^n (\ln(2))^2} = 0$$

Third differentiation:

$$\lim_{n \rightarrow \infty} \frac{6}{2^n (\ln(2))^3} = 0$$

$$n^3 \in O(2^n)$$

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Using Limit

O -Notation

If $f(n)$ and $g(n)$ are growth functions such that

$$\lim_{n \rightarrow \infty} f(n)/g(n) = \infty/\infty$$

then by **L'Hopital's Rule**

$$\lim_{n \rightarrow \infty} f(n)/g(n) = \lim_{n \rightarrow \infty} f'(n)/g'(n)$$

Example: $\lg n \in O(n)$

Converting $\lg n$ (binary) to $\ln n$ (natural), $\lg n = \ln n / \ln 2$

$$\lim_{n \rightarrow \infty} \frac{\lg n}{n} = \frac{\ln n}{(\ln 2)n} = \infty / \infty. \text{ Using L'Hopital's Rule}$$

$$\lim_{n \rightarrow \infty} \frac{1}{n} = 0$$

Therefore, $\lg n \in O(n)$

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Using Limit

Ω -Notation

If $f(n)$ and $g(n)$ are growth functions such that

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = c, \quad \text{where } 0 < c \leq \infty$$

then $f(n) \in \Omega(g(n))$.

Example(1): $7n^2 + 12n + 8 \in \Omega(n^2)$

$$\lim_{n \rightarrow \infty} \frac{7n^2 + 14n + 8}{n^2} = 7 + 14/n + 8/n^2 = 7$$

Therefore, $7n^2 + 14n + 8 \in \Omega(n^2)$

Example(2): $10n^3 + 5n + 2 \in \Omega(n^2)$

$$\lim_{n \rightarrow \infty} \frac{10n^3 + 5n + 2}{n^2} = 10n + 5/n + 2/n^2 = \infty$$

Therefore, $10n^3 + 5n + 2 \in \Omega(n^2)$

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Using Limit

Θ -Notation

If $f(n)$ and $g(n)$ are growth functions such that

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = c, \quad \text{where } 0 < c < \infty$$

then $f(n) \in \Theta(g(n))$.

Note that $c = 0$ is excluded

Example(1): $45n^3 - 3n^2 - 5n + 20 \in \Theta(n^3)$

$$\lim_{n \rightarrow \infty} \frac{45n^3 - 3n^2 - 5n + 20}{n^3} = 45 - 3/n - 5/n^2 + 20/n^3 = 45$$

Therefore, $45n^3 - 3n^2 - 5n + 20 \in \Theta(n^3)$

Example(2): $n \lg n + n + n^2 \in \Theta(n^2)$

$$\lim_{n \rightarrow \infty} \frac{n \lg n + n + n^2}{n^2} = \lg/n + 1/n + 1 = 1$$

Thus, $n \lg n + n + n^2 \in \Theta(n^2)$

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Using Limit

o -Notation

If $f(n)$ and $g(n)$ are growth functions such that

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = 0,$$

then $f(n) \in o(g(n))$.

Example: $3n^2 + 5n \in o(n^3)$

$$\lim_{n \rightarrow \infty} \frac{3n^2 + 5n}{n^3} = 3/n + 5/n^2 = 0$$

Therefore, $3n^2 + 5n \in o(n^3)$

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Using Limit

ω -Notation

If $f(n)$ and $g(n)$ are growth functions such that

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \infty,$$

then $f(n) \in \omega(g(n))$.

Example : $3n^3 + 5n^2 \in \omega(n)$

$$\lim_{n \rightarrow \infty} \frac{3n^3 + 5n^2}{n} = 3n^2 + 5n = \infty$$

Therefore, $3n^3 + 5n^2 \in \omega(n)$

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Asymptotic Notation

Summary

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \alpha$$

Notation	Definition	Limit	Bound
$f(n) = O(g(n))$	$f(n) \leq c \cdot g(n)$ for some $c > 0$	$0 \leq \alpha < \infty$	Tight upper
$f(n) = o(g(n))$	$f(n) \leq c \cdot g(n)$ for all $c > 0$	$\alpha = 0$	Loose upper
$f(n) = \Omega(g(n))$	$f(n) \geq c \cdot g(n)$ for some $c > 0$	$0 < \alpha \leq \infty$	Tight lower
$f(n) = \omega(g(n))$	$f(n) \geq c \cdot g(n)$ for all $c > 0$	$\alpha = \infty$	Loose lower
$f(n) = \theta(g(n))$	$c_1 \cdot g \leq f(n) \leq c_2 \cdot g(n)$ for some $c_1 > 0, c_2 > 0$	$0 < \alpha < \infty$	Tight, upper and lower

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Properties Asymptotic Notation

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Asymptotic Notation

Constants

- If c is a **constant** then by convention

$$O(c) \in O(1), \quad \theta(c) \in \theta(1), \quad \Omega(c) \in \Omega(1)$$

The convention implies that the running time of an algorithm which does not depend on the size of input can be expressed in any of the above ways.

- If c is **constant** then

$$O(c \cdot f(n)) = O(f(n))$$

$$\theta(c \cdot f(n)) = \theta(f(n))$$

$$\Omega(c \cdot f(n)) = \Omega(f(n))$$

The above relations imply that in a summation the **multiplier constants can be ignored**
For example, $O(1000n) = O(n)$, $\theta(7 \lg n) = \theta(\lg n)$, $\Omega(100n!) = \Omega(n!)$

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Asymptotic Notation

Θ - O - Ω Relation

- If $f(n) \in \theta(g(n))$ then

$$f(n) \in O(g(n)), \quad f(n) \in \Omega(g(n))$$

- Conversely if $f(n) \in \Omega(g(n))$ and $f(n) \in O(g(n))$ then
then $f(n) \in \theta(g(n))$

Example: Since, $n(n-1)/2 \in \theta(n^2)$, therefore

$$n(n-1)/2 \in \Omega(n^2)$$

$$n(n-1)/2 \in O(n^2)$$

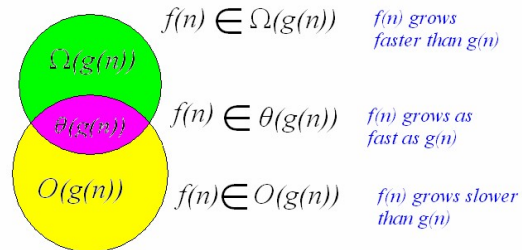
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Asymptotic Notation

Relations

If $f(n) = \Omega(g(n))$ and $f(n) = O(g(n))$ then $f(n) = \theta(g(n))$

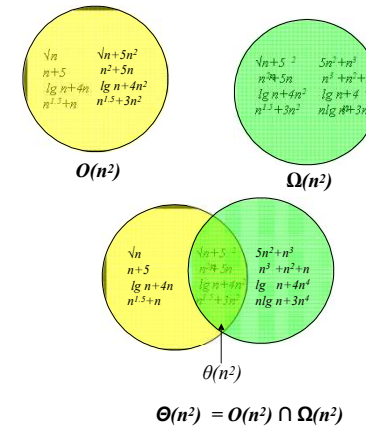
$$\theta(g(n)) = \Omega(g(n)) \cap O(g(n))$$



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Asymptotic Notation

Θ - Ω - O Relationship



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Asymptotic Notation

o - O Relation

- If $f(n) \in o(g(n))$ then

$$f(n) \in O(g(n))$$

- Converse is not true

That is, if $f(n) \in O(g(n))$, then $f(n) \notin o(g(n))$

Example(1): Since, $2^n \in o(n!)$, therefore

$$2^n \in O(n!)$$

Example (2): $n^2 + n \in O(n^2)$, but $n^2 + n \notin o(n^2)$

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Asymptotic Notation

ω - Ω Relation

- If $f(n) \in \omega(g(n))$ then

$$f(n) \in \Omega(g(n))$$

- Converse is not true

That is, if $f(n) \in \Omega(g(n))$, then $f(n) \notin \omega(g(n))$

Example(1): Since, $n! \in \omega(2^n)$ therefore

$$n! \in \Omega(2^n)$$

Example (2): $n^2 + n \in \Omega(n^2)$, but $n^2 + n \notin \omega(n^2)$

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Asymptotic Notation

Order Theorem

If $f_1(n) \in O(g_1(n))$ and $f_2(n) \in O(g_2(n))$ then

$$f_1(n) + f_2(n) \in O(\max(g_1(n), g_2(n)))$$

Proof: By definition,

$$f_1(n) \leq c_1 \cdot g_1(n) \quad \text{for } n \geq n_1$$

$$f_2(n) \leq c_2 \cdot g_2(n) \quad \text{for } n \geq n_2$$

$$\text{Let } n_0 = \max(n_1, n_2) \quad c_3 = \max(c_1, c_2)$$

$$f_1(n) \leq c_3 \cdot g_1(n) \quad \text{for } n \geq n_0$$

$$f_2(n) \leq c_3 \cdot g_2(n) \quad \text{for } n \geq n_0$$

$$f_1(n) + f_2(n) \leq c_3 \cdot g_1(n) + c_3 \cdot g_2(n) \quad \text{for } n \geq n_0$$

$$\text{Let } h(n) = \max(g_1(n), g_2(n))$$

$$f_1(n) + f_2(n) \leq 2c_3 \cdot h(n) = c \cdot h(n) \quad \text{where } c = 2c_3 \quad \text{for } n \geq n_0$$

$$f_1(n) + f_2(n) \leq c \cdot h(n) = c \cdot \max(g_1(n), g_2(n)) \quad \text{for } n \geq n_0$$

$$\text{Therefore, } f_1(n) + f_2(n) \in O(\max(g_1(n), g_2(n)))$$

Note: The above theorem also applies to θ and Ω notations

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Asymptotic Notation

Order Theorem *cont'd*

The relation

$$f_1(n) + f_2(n) \in O(\max(g_1(n), g_2(n)))$$

implies that in a summation, the **lower order growth functions can be discarded** in favor of the **highest ranking function**

Example: Consider the function $f(n)$:

$$f(n) = n + \sqrt{n} + n^{1.5} + \lg n + n \lg n + (\lg n)^2 + n^2$$

$$\text{Since, } \max(n + \sqrt{n} + n^{1.5} + \lg n + n \lg n + (\lg n)^2 + n^2) = n^2$$

$$\text{It follows that } f(n) \in O(n^2)$$

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Analysis of Summations

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Summation

Arithmetic Series

The **arithmetic series** has the summation:

$$1 + 2 + 3 + \dots + n = n(n+1)/2$$

$$\text{Let } f(n) = 1 + 2 + 3 + \dots + n = n(n+1)/2$$

$$\text{and } g(n) = n^2$$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{n(n+1)/2}{n^2} = \frac{n^2/2 + n/2}{n^2} = n/2 + 1/2$$

$$\lim_{n \rightarrow \infty} n/2 + 1/2 = 1/2, \quad \text{which is non-zero finite}$$

$$\text{Therefore, } 1 + 2 + 3 + \dots + n \in \theta(n^2)$$

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Summations

Geometric Series

The behavior of *geometric series*

$$1 + r + r^2 + \dots + r^n$$

depends on geometric ratio r . Three cases need to be considered

Case $r > 1$:

$$f(n) = 1 + r + r^2 + \dots + r^n = \frac{r^{n+1} - 1}{r - 1} \in \theta(r^n)$$

Case $r = 1$:

$$f(n) = 1 + 1 + 1 + \dots + 1 = n \in \theta(n)$$

Case $r < 1$:

$$f(n) = 1 + r + r^2 + \dots + r^n = \frac{1 - r^{n+1}}{1 - r} \in \theta(1)$$

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Summations

Logarithmic Series

The *logarithmic series* has the summation

$$\lg(1) + \lg(2) + \lg(3) + \dots + \lg(n)$$

$$\text{Let } f(n) = \lg(2) + \lg(3) + \dots + \lg(n) = \lg(2 \cdot 3 \cdot \dots \cdot n) = \lg(n!) \\ \text{and } g(n) = n \lg n$$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{\lg n!}{n \lg n} = \frac{\lg(\sqrt{(2\pi n)(n/e)^n})}{n \lg n} \quad (\text{using Stirling's Approximation})$$

$$\text{Since, } \lg(\sqrt{(2\pi n)(n/e)^n}) = (1 + \lg \pi + \lg n)/2 + n \lg n - n \lg e, \text{ therefore}$$

$$\lim_{n \rightarrow \infty} \frac{\lg(\sqrt{(2\pi n)(n/e)^n})}{n \lg n} = (1 + \lg \pi + \lg n)/(2 n \lg n) + 1 - \lg e / \lg n = 1 \quad (\text{non zero finite})$$

$$\lg(1) + \lg(2) + \lg(3) + \dots + \lg(n) \in \theta(n \lg n) \dots$$

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Summations

Harmonic Series

The *Harmonic series* has the summation

$$1 + 1/2 + 1/3 + \dots + 1/n = \lg(n) + \gamma + 1/2n - 1/12n^2 \\ \text{where } \gamma \approx 0.5772 \text{ Euler's constant}$$

$$\text{Let } f(n) = 1 + 1/2 + 1/3 + \dots + 1/n, \text{ and } g(n) = \lg n$$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \frac{\lg n + \gamma + 1/2n - 1/12n^2}{\lg n} \\ = 1 \quad (\text{non-zero finite})$$

$$\text{It follows that, } 1 + 1/2 + 1/3 + \dots + 1/n \in \lg n$$

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Summations

Polynomials

Let $p(n)$ be a polynomial of degree k ,

$$p(n) = a_0 + a_1n + a_2n^2 + \dots + a_kn^k, \text{ where } a_k > 0$$

(1) Let $g(n) = n^k$

$$\lim_{n \rightarrow \infty} \frac{p(n)}{g(n)} = \frac{a_0 + a_1n + a_2n^2 + \dots + a_kn^k}{n^k} = a_k > 0$$

$$\text{Thus, } p(n) \in \theta(n^k), \quad p(n) \in O(n^k), \text{ and } p(n) \in \Omega(n^k)$$

(2) Let $g(n) = n^{k+1}$

$$\lim_{n \rightarrow \infty} \frac{p(n)}{g(n)} = \frac{a_0 + a_1n + a_2n^2 + \dots + a_kn^k}{n^{k+1}} = 0$$

$$\text{Thus, } p(n) \in o(n^{k+1}), \text{ and } p(n) \in O(n^{k+1})$$

(2) Let $g(n) = n^{k-1}$

$$\lim_{n \rightarrow \infty} \frac{p(n)}{g(n)} = \frac{a_0 + a_1n + a_2n^2 + \dots + a_kn^k}{n^{k-1}} = \infty$$

$$\text{Thus, } p(n) \in \omega(n^{k-1}), \text{ and } p(n) \in \Omega(n^{k-1})$$

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